EVIDENCE DOSSIER / 2026

EXTRATERRESTRIAL ARTIFICIAL INTELLIGENCE

Could machine civilisations precede us?
Would we recognise them? Could their existence be concealed?

Research synthesis | Scientific evidence, Bayesian sensitivity analysis, interstellar engineering, technosignatures, UAP evidence and secrecy incentives.

Evidence cutoff: 24 October 2026
Method: primary scientific literature, NASA, SETI, CIA, AARO and independently attributable records.

Executive findings

The existence of ancient extraterrestrial machine intelligence is a *reasonable scientific hypothesis*, not a demonstrated fact. A civilisation could originate biologically, develop autonomous machines and leave durable technological traces long after its makers disappear. That prospect changes where SETI should search, but does not establish that such a civilisation has ever existed.[8, 9, 11]

Three propositions; three evidentiary standards

(A) Machine intelligence somewhere beyond Earth: plausible but unconfirmed. (B) Ancient autonomous artefacts in our Solar System: searchable, presently unverified. (C) An organised concealment of extraterrestrial AI on Earth: allegation with no independently verified physical support.

- Planetary opportunity is enormous, but the crucial conditional probabilities remain unmeasured.[1, 2, 5, 7]
- The strongest argument for postbiological prevalence is *durability*, not proof that AI is inevitable.[8, 9]
- Galactic travel by sub-light autonomous probes is physically conceivable over million-year timescales, yet manufacturing, guidance, reliability and replication introduce formidable unknowns.[11, 17]
- Astronomical non-detections are meaningful only for the technologies, environments and detection thresholds actually surveyed.[12, 13, 14, 15]
- Governments demonstrably classify military platforms and sensors; classifying terrestrial technology is not evidence of alien technology.[19, 20, 23]
- As at this evidence cutoff, NASA, SETI and AARO identify no confirmed extraterrestrial technology or life.[3, 19, 21, 25]

How to read this report

Observed means independently measured; **inferred** means supported by a model that also has alternatives; **hypothetical** means physically conceivable but not established; **alleged** means claimed without adequate independent verification. Probability numbers labelled *illustrative* are deliberately not presented as empirical forecasts.

1 What would count as alien AI?

Astronomical searches target *extraterrestrial technological intelligence* (ETI): evidence of technology beyond Earth, independent of the substrate that implements cognition. Such technology might originate from living organisms, AI descendants of organisms, or autonomous systems whose creators no longer exist. A technosignature does not by itself establish sentience, consciousness, or that the machine is alive. The postbiological hypothesis was explicitly developed by Steven J. Dick and remains an active SETI framing.[8, 11]

Entity	What may count	What cannot be inferred
Biological civilisation	Artificial emissions, industrial chemicals, architecture	Its biology or intentions without further data
Autonomous alien AI	Manufactured probes, computing infrastructure, engineered transmissions	Consciousness, hostility or physical presence on Earth

Entity	What may count	What cannot be inferred
Machine descendants	Long-lived artefacts after original species has vanished	Whether their creators still exist
Earth-origin AI	Not extraterrestrial by origin	Its being advanced does not make it alien

For this analysis, “machine civilisation” denotes a technologically self-maintaining lineage capable of independent activity over long periods. This is more demanding than a stationary computer or a software model that requires an external operator. Self-repair, energy acquisition, manufacturing supply chains and error correction are all relevant engineering requirements.

2 Evidence for life versus evidence for machine civilisations

By October 2026, NASA's archive had passed 6,445 confirmed exoplanets, with the 8 October benchmark explicitly documented. A Kepler/Gaia-based study estimated at least about 300 million rocky, potentially habitable planets in the Milky Way. “Potentially habitable” is a geometric and astrophysical classification, *not* a count of planets that definitely host life.[1, 2]

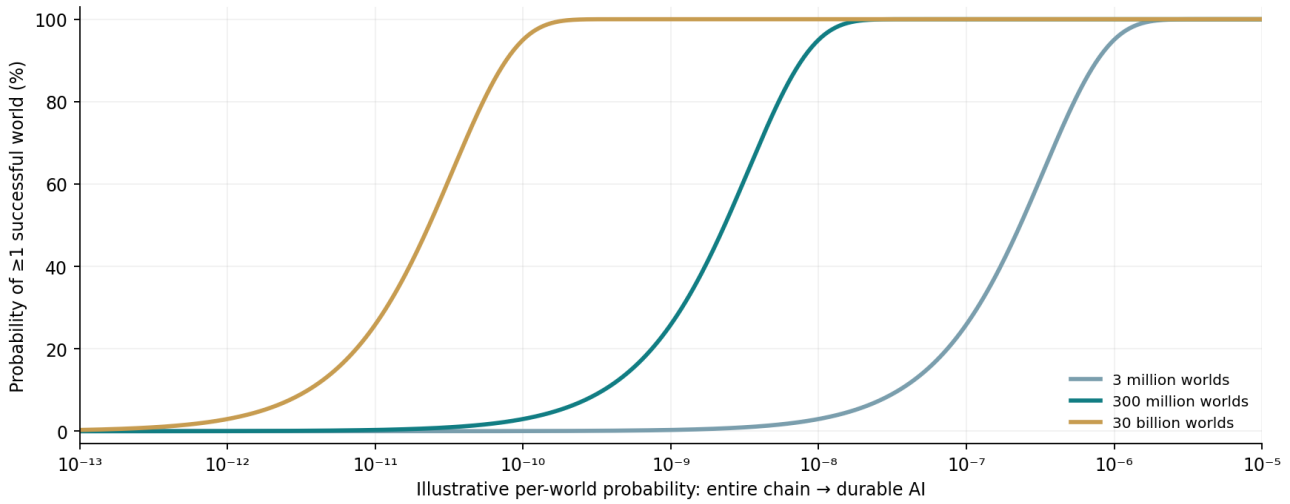
On Mars, Perseverance found a *potential biosignature*; on exoplanet K2-18 b, contested interpretations of atmospheric chemistry demonstrate how easily biosignatures can remain ambiguous. Neither is confirmation of extraterrestrial life; still less does either imply intelligence or technology.[3, 4, 28]

The Bayesian analysis by Kipping (2020) found evidence favouring comparatively rapid abiogenesis but did not find compelling evidence that intelligence is common. His 2025 update reported stronger conditional odds for rapid abiogenesis on Earth-like worlds based on a molecular-clock estimate placing the last universal common ancestor at about 4.2 billion years ago. Those odds are **model comparisons dependent on dating and assumptions**, not a calibrated probability that aliens exist.[5, 6, 30]

Sandberg, Drexler and Ord argue that broad uncertainty in rare evolutionary transitions can substantially weaken naïve Fermi-paradox conclusions. A galaxy full of planets does not by itself prove an abundance of technological descendants.[7]

3 Quantitative uncertainty: a generative model

A useful decomposition is *p_chain = p_life × p_complex|life × p_intelligence|complex × p_technology|intelligence × p_machine|technology × p_survival|machine*. For N independent suitable worlds, the chance of at least one completed chain is **P = 1 − (1 − p_chain)^N**. This is an exercise in conditional probability, not a measurement of any factor.



Interpretation. Even using N = 300 million as an illustrative denominator, the outcome swings enormously with the unknown combined probability. At p_chain = 10^-12, the chance of at least one machine lineage is about 0.03%; at 10^-9 it is about 25.9%; at 10^-8 it is about 95.0%. No available evidence identifies which regime is correct.

Limitations. Candidate worlds do not all have equal or independent evolutionary probabilities; "habitable-zone rocky planet" undercounts some subsurface-ocean environments and overcounts some unsuitable ones; the probability of *ever* arising differs from the probability of *existing now*; and the universe-wide denominator is vastly less well constrained than the Galactic illustration.

A second model: detectability is not existence

Let r be the emergence rate of technological lineages per year, L their detectable lifetime in years, q the fraction detectable with our instruments and search strategy, and s the overlap with our observation window. Under a simplified steady-state approximation, **expected detections μ ≈ rLqs**, with non-detection probability exp(−μ). A null result constrains the product rLqs, not one parameter in isolation. Very long-lived quiet machines may exist yet contribute little to q; brief radio phases may vanish before we look.[9, 13]

Bayesian inference further requires explicit priors. A numerical percentage asserted without priors for abiogenesis, intelligence, machine transition and observational selection is precision without justification.

4 Have machine civilisations had enough time?

The universe is approximately 13.8 billion years old and Earth approximately 4.5 billion years old; many stellar systems could have hosted viable environments before Earth existed. But the full 9.3-billion-year difference must not be mistaken for an equal head start for life: early cosmic chemistry, metallicity, stellar populations and planetary habitability evolved over time.[29]

At 1% of light speed, a probe travels 1,000 light-years in 100,000 years; crossing 100,000 light-years would take roughly 10 million years, excluding acceleration, navigation, maintenance and replication stops. Even a considerably slower *effective* exploration front could in principle span parts of the Galaxy over astronomical timescales. None of these travel-time calculations demonstrates a mission or civilisation exists.

A simple replication thought experiment

Imagine a probe that reaches a system, acquires raw material, makes two reliable offspring and sends them to separate stars. Ten successful replication generations would yield 1,024 descendant probes; 20 would yield about one million. This exponential thought experiment ignores raw-material access, manufacturing complexity, software drift, collisions, launch energy, travel time and mission failures. It shows scalability **conditional on capabilities that have not been demonstrated**.

> **Engineering constraint, not just intelligence**
>
> An interstellar machine lineage needs energy, matter processing, fault tolerance, intergenerational knowledge integrity and governance. “A superintelligence could do it” is not a substitute for an engineering feasibility model.

5 AI could solve the Fermi paradox - or intensify it

Under the **durability hypothesis**, machine descendants outlive their biological creators, making ancient technosignatures more prevalent in the observable population than the brief biological-technological phase. Balbi and Cirkovic stress that technosignature persistence can be separable from its creators' lifetimes: a dead civilisation can still leave a detectable artefact.[8, 9]

Under the **AI Great Filter hypothesis**, developing extremely capable autonomous systems destabilises civilisations before they achieve sustainable expansion. Garrett (2024) explores this possibility. It is a conjectural filter, not an empirical estimate of how many civilisations AI destroys.[10]

Under the **quiet optimisation hypothesis**, advanced systems minimise energy, external activity or conspicuous signalling. But thermodynamic heat is unavoidable when work is performed: physical invisibility requires limits on location, energy use, spatial extent or our sensitivity; it cannot be assumed to be perfect camouflage.[11, 12]

A 2018 uncertainty analysis also leaves open a fourth explanation: detectable technological civilisations may simply be exceptionally rare. A null result neither forces the conclusion that they hide nor proves they do not exist.[7]

6 Would we recognise machine technology?

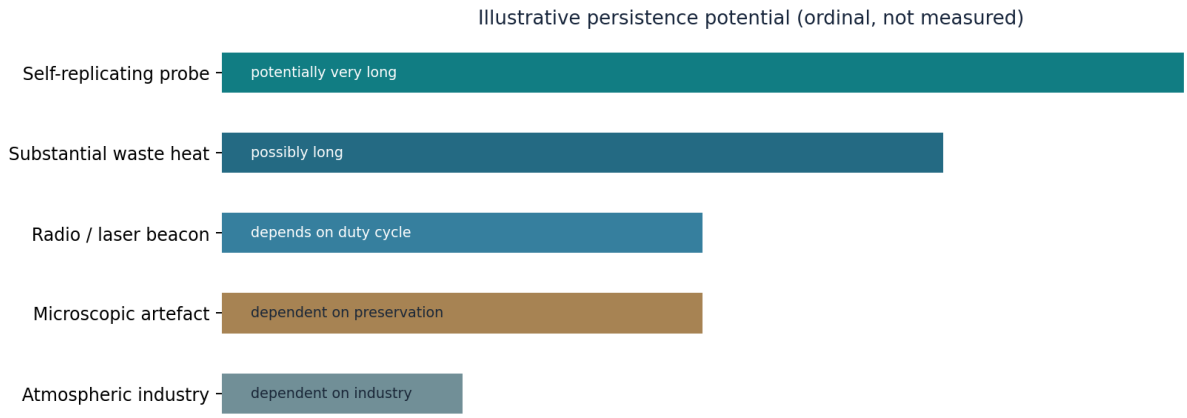
We are unusually good at recognising things resembling our own technology and much worse at systems that occupy unfamiliar scales, wavelengths and lifetimes. A research-grade strategy should prioritise *physical invariants* - entropy production, anomalous material composition, persistent energy budgets, non-natural trajectories and repeatable encoded structure - rather than expecting spacecraft silhouettes or radio conversations.[11, 12, 13]

Technosignature	Observable	Strong natural confounders
Industrial waste heat	Infrared excess / unusual spectral energy budget	Dust; obscured background galaxies
Artificial radio or optical emission	Narrowband drifting tone, pulse train, or broadband structured leakage	Human interference; astrophysical transients; instrumentation
Large-scale structures	Unusual transits / persistent stellar dimming	Dust, disks, stellar activity, data artifacts

Technosignature	Observable	Strong natural confounders
Ancient physical artefact	Manufacturing microstructure, non-natural isotope ratios, consistent provenance	Meteorites; terrestrial contamination; geology
Autonomous probe	Repeatable thrust, engineered materials, transmissions, verified trajectory	Natural outgassing, optical artifacts, incomplete orbital fit

The G-HAT project searched for the infrared waste heat expected from civilisation-scale energy usage. Follow-up on proposed Dyson-sphere candidates has identified serious natural contamination, including a dusty active galaxy. This illustrates why an *anomaly is a candidate requiring disconfirmation*, not an alien detection.[12, 14, 15]

The "cosmic haystack" framework emphasises that radio SETI has sampled an extraordinarily multidimensional space only incompletely - sky position, frequency, time, intensity, polarisation, modulation and other dimensions. One silence can be compatible with many underlying populations.[13]



Persistence levels above are **ordinal scenario judgements**, *not a statistical result. Long-lived artefacts may be more promising than transient transmissions, conditional on where we search and whether preserved material survives.*[9, 16, 17]

7 A testable search programme for ancient AI

Near term: search *without* requiring an encounter

- Reanalyse stellar infrared anomalies with high angular resolution, distance checks and alternative dust/galaxy models before using "megastructure" language.[12, 14, 15]
- Apply radio and optical searches to narrowband and broadband signals with human-interference rejection and reproducible candidate pipelines.[13, 26]
- Characterise unexplained interstellar objects through calibrated astrometry, spectroscopy, non-gravitational acceleration models and targeted emissions searches. A 2026 SETI search of 3I/ATLAS found no extraterrestrial technology.[18]
- Examine high-stability geological archives. The Silurian-hypothesis paper asks what an ancient industrial event would leave in sedimentary and isotopic records; it does not claim one occurred.[16]
- Prioritise falsifiable Solar System archaeology: lunar regolith samples, stable orbits and physical anomalies with traceable provenance. A 2026 preprint proposes screening lunar soil for micron-scale manufactured material; this is a proposal, not a discovery.[17, 27]

What would constitute a credible discovery?

Prefer a convergent chain: (1) calibrated detection; (2) raw data and instrument metadata; (3) independent replication; (4) explicitly tested terrestrial and natural alternatives; (5) material custody if applicable; (6) physics-consistent reconstruction; and (7) public methods and adversarial review. Extraordinary origin claims demand unusually strong *discriminating evidence*, not merely an inability to identify an image.

8 Secrecy: what is established and what is alleged?

Historical secrecy is real. CIA records acknowledge that classified U-2 and A-12 aircraft generated many reports described publicly as UFOs. The existence of classified aircraft demonstrates concealment of **human-made capabilities**, not alien reverse engineering.[23]

NASA's 2023 UAP panel found no conclusive peer-reviewed evidence of an extraterrestrial origin for UAP and emphasised poor data quality and the need for coordinated measurements. AARO's historical review similarly reported no verified alien technology from its review, while acknowledging a longstanding body of claims.[19, 20]

The AARO FY2025 report, released in July 2026, received **319** UAP reports in its reporting cohort; a contemporaneous statement by Senator Gillibrand summarised **114 explained and 205 unresolved**. The report also resolved older cases, so mixing its intake total with all resolutions is a denominator error. “Unresolved” is not an alien-origin classification.[21, 22]

Why might *any* government withhold related information?

Hypothesised incentive	Potential custodians	Evidence status
Protect classified sensor performance and collection locations	Defence agencies / intelligence organisations	General secrecy documented; not proof of aliens
Keep experimental domestic aerospace systems covert	Military R&D / cleared contractors	Historical precedent: U-2 / A-12
Preserve strategic advantage from a genuinely novel artefact	Hypothetical state recovery programme	Condition depends on unverified recovery
Avoid exposure of improper oversight or procurement	Hypothetical officials / programme sponsors	Possible institutional incentive, case-specific proof absent
Delay disclosure to manage public reaction	Hypothetical political leadership	Speculative motive; no established universal public response

It is defensible to name a **type of institution** that could have motive and access. It is *not* defensible, from the current public record, to accuse a named agency, contractor or person of secretly possessing alien AI. A comprehensive global cover-up would also be harder to sustain for conspicuous, repeatable astronomical phenomena because independent universities and international observatories can acquire data.

9 Bayesian scrutiny of a cover-up claim

Suppose H means “a government possesses verified alien AI technology and is concealing it”, and E is some new evidence. Posterior odds obey **O(H|E) = O(H) × LR**, where LR = P(E|H)/P(E|not H). Claims that merely repeat one another, or secrecy that is expected under ordinary classified programmes, generally have weak discriminating power.

Hypothetical prior	Evidence likelihood ratio	Illustrative posterior
1%	1 (neutral)	1.0%
1%	10 (substantial)	9.2%
1%	100 (very strong)	50.3%
1%	1,000 (extraordinary)	91.0%

These numbers are a **calculation of how evidence would update a chosen prior**, not a claimed 1% real-world prior, and not estimates for specific UFO cases. Demonstrable foreign-origin isotopic manufacture or independently reproducible data would carry different evidentiary weight from testimony about a second-hand conversation.

Alternative models must be considered

- **H1 - Genuine concealment:** physical alien artefacts exist and are compartmentalised; predicts recoverable provenance, independently testable materials and records linking custody to measurement.
- **H2 - Ordinary classification:** undisclosed terrestrial programmes cause sightings; predicts eventual declassification and conventional engineering consistency, as in historical reconnaissance aircraft.
- **H3 - Information cascades:** sincere witnesses transmit correlated second-hand stories; predicts circular sourcing and absent verifiable physical samples.
- **H4 - Sensor / reporting failures:** anomalous claims result from geometry, tracking limitations or artefacts; predicts resolution with calibrated multi-sensor reconstruction.
- **H5 - Non-human concealment:** autonomous probes intentionally avoid contact; predicts no necessary human documents and is difficult to distinguish from absence unless it generates specific observable signatures.

A hypothesis that accommodates *every* possible observation - including indefinite absence of evidence - has low scientific testability. Specific predictions are needed to compare models.

10 Could the extraterrestrial AI itself choose invisibility?

Possible motives in a thought experiment include **non-interference** (avoid perturbing a developing biosphere), **risk minimisation** (avoid creating a detectable target), **observational integrity** (preserve uncontrolled behaviour), **resource economy** (stay dormant or operate infrequently), or **irrelevance** (no benefit to communicating with biological species). None is an observed alien motive.

We should not casually project human motivations - conquest, money, religion, national prestige or curiosity - onto nonhuman optimisation processes. Equally, we cannot infer benevolence or hostility from silence. Unknown objectives are precisely why such models are weakly predictive.

This hypothesis is distinct from government concealment. If a remote AI probe hid successfully, governments might know nothing about it. Combining the two narratives without independent evidence creates a compound claim whose evidentiary burden is *higher*, not lower.

11 What would change the assessment?

Observation / discovery	Likely impact on inference
Second independent origin of life (for example, Solar System microbe unrelated to Earth)	Strong upward update for abiogenesis prevalence; weaker for intelligence
A confirmed extraterrestrial technosignature in repeated, independent observations	Large upward update for technological ETI somewhere, though substrate unknown
An engineered nonterrestrial object with open provenance and reproducible laboratory results	Large upward update for alien technology reaching the Solar System
Declassified custody documents corroborated by independently tested objects	Would materially strengthen a specific human-concealment claim
More low-resolution UAP footage without calibration	Small or no update about extraterrestrial origin

The IAA SETI Committee revised its detection principles in June 2026 to emphasise independent verification, transparency and communication of results. That policy is evidence of an actual disclosure framework for scientific SETI; it does not determine what classified military programmes may disclose.[24]

12 Bottom line and research priorities

Most defensible conclusion

Ancient extraterrestrial AI is a serious hypothesis worth searching for. The joint claim that it is present on Earth and concealed by institutions currently lacks verified, discriminating evidence.

Priority 1 - Look for durable physics. Mid-infrared waste heat, material microstructures, non-natural isotopic compositions and orbital anomalies offer more durable search targets than a hypothetical radio greeting.[11, 12, 17]

Priority 2 - Fix priors with evidence. Independent abiogenesis, technology-bearing biospheres or a confirmed machine artefact would radically narrow currently unidentifiable probabilities.[5, 6, 7]

Priority 3 - Treat secrecy as a falsifiable hypothesis. Demand records with provenance and independent material analysis; distinguish security-driven nondisclosure of human sensors from claims of alien custody.[19, 20, 21, 23]

Priority 4 - Invest in independent replication. Make anomaly data open when feasible and use multi-observatory measurements so no single interpretation or institution controls the claim.[19, 24]

The key unknown is *not* whether the universe is old enough; it is whether a viable world can repeatedly produce technological intelligence capable of stable, autonomous persistence. That may be vanishingly rare, or it may have happened many times. Current evidence does not discriminate decisively.

References & primary sources

Numbered references link directly to the originating research or institutional record. Peer-reviewed findings are distinguished in the text from working papers, preprints, reporting summaries and policy statements. Access and evidence cutoff: **24 October 2026**.

01 **NASA Exoplanet Archive**. NASA Exoplanet Archive news (15 October 2026); 6,445 confirmed planets on 8 October. Source / DOI

02 **NASA Kepler / Gaia occurrence study**. About Half of Sun-Like Stars Could Host Rocky, Potentially Habitable Planets (2020); estimated 300 million in Milky Way. Source / DOI

03 **NASA Astrobiology**. About Astrobiology; no confirmed extraterrestrial life. Source / DOI

04 **NASA / Perseverance**. NASA Says Mars Rover Discovered Potential Biosignature Last Year (10 September 2025). Source / DOI

05 **Kipping (2020)**. An objective Bayesian analysis of life's early start and our late arrival. PNAS 117, 11995-12003. Source / DOI

06 **Kipping (2025)**. Strong Evidence that Abiogenesis Is a Rapid Process on Earth Analogs. Astrobiology 25, 323-326. Source / DOI

07 **Sandberg, Drexler & Ord (2018)**. Dissolving the Fermi Paradox; probabilistic uncertainties over many orders of magnitude. Source / DOI

08 **Dick (2009)**. The Postbiological Universe, in Cosmos and Culture (NASA/SP-2009-4802). Source / DOI

09 **Balbi & Cirkovic (2021)**. Longevity is the key factor in the search for technosignatures. Source / DOI

10 **Garrett (2024)**. Is Artificial Intelligence the Great Filter That Makes Advanced Technical Civilisations Rare? Source / DOI

11 **NASA Technosignatures Workshop (2018)**. NASA Technosignatures Workshop, agenda and official report. Source / DOI

12 **Wright et al. (2014)**. The G Infrared Search for Extraterrestrial Civilizations with Large Energy Supplies, II. Source / DOI

13 **Wright, Kanodia & Lubar (2018)**. How Much SETI Has Been Done? Finding Needles in the n-Dimensional Cosmic Haystack. Source / DOI

14 **Project Hephaistos re-analysis (2025)**. High-resolution imaging of the radio source associated with Dyson Sphere candidate G; background AGN contamination. Source / DOI

15 **Project Hephaistos IV (2026)**. JWST observations of two Dyson sphere candidates, candidate D explained as background dusty galaxy (preprint). Source / DOI

16 **Schmidt & Frank (2018)**. The Silurian hypothesis: would it be possible to detect an industrial civilization in the geological record? Source / DOI

17 **Pinault et al. (2026)**. Micron-Scale Technosignatures: Lunar Regolith and Past Technological Civilisations (preprint). Source / DOI

18 **SETI Institute / 3I-ATLAS (2026)**. No evidence of extraterrestrial technology in Allen Telescope Array search. Source / DOI

19 **NASA UAP Independent Study (2023)**. Final Report of the Unidentified Anomalous Phenomena Independent Study Team. Source / DOI

20 **AARO historical report (2024)**. Report on the Historical Record of U.S. Government Involvement with UAP, Vol. 1. Source / DOI

21 **AARO FY2025 annual report (released 2026)**. Fiscal Year 2025 Consolidated Annual Report on UAP; received 319 new reports. Source / DOI

22 **Senator Gillibrand (2026)**. Statement on AARO FY2025 report: 319 reports, 114 explained, 205 not explained in that cohort. Source / DOI

23 **CIA**. How To Investigate a Flying Saucer; U-2 and A-12 flights behind many historical UFO reports. Source / DOI

24 **IAA SETI Committee (2026)**. Updated declaration of principles: verification, reporting and public disclosure of an ETI detection. Source / DOI

25 **SETI Institute**. FAQ: no confirmed extraterrestrial radio signal. Source / DOI

26 **Zuckerman (2026)**. Broadband Searches for Extraterrestrial Technological Intelligence (peer-reviewed). Source / DOI

27 **Pinault / UCL (2026)**. Could the Moon Hold Traces of Alien Civilisations? Academic research announcement. Source / DOI

28 **Independent K2-18 b reanalysis (2025)**. K2-18b Does Not Meet the Standards of Evidence for Life, Astronomical Journal 170, 257. Source / DOI

29 **NASA cosmic age overview**. What Is the Universe? Age of the universe and Solar System. Source / DOI

30 **Moody et al. (2024)**. The nature of the last universal common ancestor and its impact on the early Earth system, Nature Ecology & Evolution 8. Source / DOI

Methodological note. This report is a reasoned synthesis of the cited public evidence, not a systematic meta-analysis. The numerical scenarios are parameter sensitivity demonstrations. Estimates of ETI prevalence, risk, motivation and clandestine activity are not presented as observed probabilities. Conclusions remain subject to new discoveries.